



Near-future oceanic CO₂ delays development and growth in early-stage larvae of the endemic New Zealand sea urchin, *Evechinus chloroticus*

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Abstract

Under the ocean acidification (OA) conditions predicted for 2100, the larval stages of temperate sea urchins are smaller, with reduced and abnormal skeleton and changes in metabolic rate. Here, we measure the impact of near-future and long-term predictions of atmospheric pCO₂ levels on the early development of the echinometrid sea urchin *Evechinus chloroticus* using single male:female crosses, effect sizes and a within-study meta-analysis. Using the developmental signpost of the 16-cell stage, we show a mean delay of 5.11 min at pH 7.7 and 11.85 min at pH 7.5, which may have flow-on effects to later embryo and larval stages. Echinopluteus larvae raised in OA conditions (pH 7.7, 7.5) were ~5–10% smaller in body length and 10–12% smaller in arm length than controls. Metabolic rate was highly variable between single male:female crosses—increasing in some crosses, decreasing in others—with a non-significant effect size in the meta-analysis. Preliminary experiments suggest that metabolic rate changes may be impacted by loss of mitochondrial function at low pH. Single male:female crosses showed variable OA responses in all measurements, suggesting that when assessing the population-level impacts of OA on early development of sea urchins there is a need to include high levels of biological replication.

Introduction

Anthropogenic atmospheric CO₂ has reduced the pH of the world's oceans by 0.1 units since the industrial age and is predicted to decrease by another 0.3–0.4 pH units by 2100 (IPCC 2014). This reduction in pH represents a 30% elevation in H⁺ concentration; hence this process is known as ocean acidification (OA, Orr et al. 2005; Doney et al. 2009). Lower pH also decreases carbonate concentrations, which

can impact calcification in organisms with calcium carbonate skeletons (e.g., coccoliths, coralline algae, corals, molluscs, echinoderms) while also affecting organism growth, physiology (metabolism, acid–base regulation, stress response), and behavior (Pörtner et al. 2004; Fabry et al. 2008; Doney et al. 2009; Hofmann et al. 2010; Kroeker et al. 2010; Przeslawski et al. 2015; Byrne and Fitzer 2019; Strader et al. 2020; Thomas et al. 2020).

Echinoderms, which have a calcium carbonate skeleton, were identified as a phylum that might be particularly vulnerable to OA and were the focus of much early research (19 studies reviewed in Dupont et al. 2010; recent reviews: Wittmann and Pörtner 2013; Sokolova et al. 2016; Byrne and Fitzer 2019; Byrne and Hernandez 2020). Sea urchin embryos, when cultured in near-future OA conditions, develop into larvae with shorter skeletal rods and less total skeleton (“stunting”; reviews: Byrne et al. 2013, 2018; Byrne and Hernandez 2020) and/or delayed development, resulting in embryos and larvae that are at a “younger” ontogenetic stage than controls (e.g., Martin et al. 2011; Stumpp et al. 2011a; Place and Smith 2012; Zhan et al. 2016). Further, sea urchins reared to metamorphosis in OA conditions are smaller at settlement (Wangensteen et al. 2013), and juveniles grow slower in their first five months (Dworjanyn and

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Byrne 2018). However, as noted in both early and recent reviews, the impact of OA is species-specific (Dupont et al. 2010; Byrne 2012; Byrne and Fitzer 2019; Byrne and Hernandez 2020), necessitating a comprehensive geographic and phylogenetic approach to the study of developmental delay and “stunting” in sea urchins.

The endemic New Zealand echinometrid *Evechinus chloroticus* (Valenciennes 1846) is geographically distributed throughout New Zealand, generally found at depths of 12–14 m, although it can also occur in the intertidal zone (Barker 2020). Two OA studies have been conducted on this species to date. Clark et al. (2009) measured the impacts of pH 8.1 and 7.7 on the larval skeleton at 13 days post-fertilization, but larvae were not cultured in OA conditions until after the hatched blastula stage (Clark et al. 2009; Byrne et al. 2013), which occurs ~13–14 h after fertilization (Sewell et al. 2004). Espinel-Velasco et al. (2020) tested the impacts of OA on settlement success in a range of pH conditions (8.0 to 7.0); however, the larvae were grown to competency in control conditions and were exposed to OA conditions for only a short period (24–48 h) during the process of settlement.

To understand the impact of OA on early development on *E. chloroticus* we focused on three aspects of physiology—development time, larval morphology, respiration—during the first four days of development. With increasing recognition of the importance of individual responses to climate change (e.g., Pistevos et al. 2011; Schlegel et al. 2011; Guscetti et al. 2019) and quantitative evidence for high heritability of embryo and larval traits in sea urchins in response to OA (e.g., Sunday et al. 2011; Kelly et al. 2013; Foo et al. 2014) we used single male:female crosses from sea urchins collected from a single population. A within-study meta-analysis using effect sizes (Garamszegi 2006; Nakagawa and Hauber 2011; Nakagawa and Santos 2012) was used to test for the impact of OA using the single male:female crosses as individual studies. Within-study meta-analysis, demonstrating the existence of an effect using a series of experiments, can be used to detect smaller treatment effects than a null hypothesis significance testing approach (Nakagawa and Hauber 2011), and also allows investigation of differences in OA responses between genetic crosses and the potential for adaptation in this population.

Our research on developmental time (Part 1) sought to determine if there are delays during embryonic cleavage in sea urchins in OA conditions by considering beyond the first cleavage (Place and Smith 2012; Limatola et al. 2020), using near-future pH conditions, and with a higher sampling frequency (<30 min, Zhan et al. 2016) to detect subtle differences in development time between male:female crosses in OA conditions.

Part 2, on the impacts of OA on larval morphology in *E. chloroticus*, used the early 4-arm pluteus stage (48 h)

in unfed cultures to avoid confounding OA effects on arm length with phenotypic plasticity due to the presence and concentration of algal food (Strathmann et al. 1992; McAlister and Miner 2018), and/or any secondary effects of OA on swimming and feeding (Chan et al. 2011, 2015). The 48 h time-point was specifically chosen as fed and starved *E. chloroticus* larvae at this time have a similar morphology (Sewell et al. 2004), and quantity of energetic lipids (Sewell, 2005), so any observed changes in arm length in OA conditions are not confounded by energetic constraints or the presence/absence of food.

Finally, in Part 3 we measured organismal metabolic rate as a measure of whole animal performance and fitness in OA conditions. In a preliminary study we also tested, using measures of mitochondrial function, whether there was evidence for disruption to the electron transport chain, which has been suggested from gene expression studies in sea urchins in OA conditions (e.g., Todgham and Hofmann 2009; Evans and Watson-Wynn 2014).

Together these experiments show generally negative effects of OA on embryo and larval development of *E. chloroticus*, but often with a highly variable physiological response between single male:female crosses of sea urchins collected from a single population.

Materials and methods

Study of the effect of OA on early development of *Evechinus chloroticus* proceeded in three parts during the austral summers of 2012 and 2013, on a single population with the same thermal history. Part 1, early development to the 16-cell stage ~2–6 h post fertilization used five single male:female crosses (crosses A to E) from sea urchins collected in January and February 2012; Part 2, larval morphometrics at 48 h, used thirteen single male:female crosses (crosses F to R) from sea urchins collected in February 2012; and Part 3, metabolic rate at 72 and 96 h, used seven single male:female crosses from sea urchins collected in February and March 2013 (crosses S to Y). In each part the species OA response was based on the independent full-sibling crosses as the unit of biological replication.

Urchin collections and spawning

Adult *Evechinus chloroticus* were collected by free-diving from Matheson's Bay (36° 18.10' S, 174° 48.65' E) in the Hauraki Gulf, New Zealand. Urchins were transported by car in chilled containers to the University of Auckland (ca 100 km) and held in aerated static tanks (280 L) in a temperature-controlled room (18 °C) and a 12:12 light:dark cycle until use. The seawater in all experiments was from the waters of the Waitemata Harbor, Auckland, New Zealand,

prefiltered through sand media filters at Kelly Tarlton's Sea Life Aquarium, and transported to the University of Auckland in 20 L carboys. Immediately before use this water was filtered through a series of two nested 1 μm filter bags to produce 1 μm filtered seawater (FSW).

Spawning was induced by intracoelomic injection of 1–3 ml of 0.55 M KCl solution (amount based on animal size). Eggs from individual females were collected over beakers of FSW, and “dry” sperm from individual males collected using a glass pipette and stored in a 1.7 ml Eppendorf tube on ice for up to 1 h before use. Eggs from each female were fertilized in a 250 or 500 ml beaker with a dilute sperm suspension (20 μl of “dry sperm” to 15 ml of FSW) from a single male in FSW; beakers from each male:female cross were kept on a shaker table for 30–60 min. Male:female crosses with fertilization greater than 80% were washed three times in FSW (80% water change) to remove excess sperm, and transferred to the culture buckets with experimental water (see below) within ~30 min to 1 h of fertilization.

OA Treatments and seawater chemistry

OA treatment levels were based on near-future projections of pCO_2 concentrations: the reduced Representative Concentration Pathway (RCP) 4.5 at 2100 (range 500–650 μatm), and business as usual emission scenarios of RCP 8.5 at 2100 (range 851–1370 μatm) and RCP 8.5 at 2150 (range 1371–2900 μatm ; IPCC 2014). To maintain constant pCO_2 conditions in our central city laboratory we used CO_2 -free air (as in Fangue et al. 2010) blended with instrument grade CO_2 (99.98%, BOC Gases, Auckland) with three mass-flow controllers (SmartTrak, Sierra, Monterey, CA, USA), resulting in a control and two OA treatments per experimental part.

During all culture experiments, temperature and salinity were monitored daily using a handheld salinity/temperature meter (YSI 30, YSI Inc., Ohio) and samples collected from each culture bucket for spectrophotometric determination of pH_{total} . Seawater pH samples were analyzed within 2 h of collection using m-cresol purple (Sigma) based on SOP6b (Dickson et al. 2007) as modified for use in a 1 cm cell by Fangue et al. (2010). Samples were calibrated using TRIS standard (Dickson et al. 2007). Total alkalinity was measured throughout the experiments using mercuric chloride poisoned water samples (100 μl of 0.02% mercuric chloride stock solution added to 500 ml seawater). Open-cell potentiometric TA measurements were performed using a Mettler-Toledo T50 automated titrator based on SOP3b (Dickson et al. 2007) and Fangue et al. (2010) with the accuracy of TA values confirmed using Scripps certified reference seawater (Batch 108 with $\text{TA} = 2218 \mu\text{mol/kg}$). The carbonate system parameters (pCO_2 , Ω_{a} and Ω_{c}) were calculated from these

measurements using CO2SYS v. 2.1 (Pierrot et al. 2006) using the dissociation constants from Mehrbach et al. (1973) as refitted by Dickson and Millero (1987).

Larval rearing at elevated pCO_2

Embryos and larvae were reared in static 10 (Part 1) or 20 L (Part 2, Part 3) plastic buckets of FSW with a central air-lift column modified from the flow-through design of Russell (2000). The air-lift column was constructed from 32 mm diameter PVC pipe, open at the top, with 25 μm Nitex mesh on the lower end, and held vertical in the buckets by insertion through the plastic lid. Water was drawn into the air-lift column by the internal airstone and recirculated into the buckets through 4 upward angled small plastic spigots constructed from irrigation connectors that were inserted equal distance around the circumference (at 90° to each other), and located below the water surface (see Russell 2000). Pre-mixed gases at the experimental pCO_2 were delivered to the standpipe through the central airstone and equilibrated for a minimum of 16 h before addition of the fertilized embryos (100,000 per bucket, ca 5 ml^{-1}). Three 20 L buckets were prepared from each cross—one each for Control and the two pCO_2 treatments—so that full siblings from a single cross were cultured in the three experimental conditions. Cultures were maintained in the same temperature-controlled room as for the collected adults (see above) and water changed in experiments > 48 h (Part 3) by reverse filtration of 50% of the volume, and replacement with pre-equilibrated seawater at the appropriate pCO_2/pH .

Part 1: Development time to 16-cell stage

Offspring of five single male:female Crosses (A, B, C, D, E) were reared in 10 L buckets equilibrated to emission scenarios RCP 8.5 at 2100 and RCP 8.5 at 2150. We assessed the time of development for the transition from the 4-cell to the 8-cell and 8-cell to 16-cell stages (~2–6 h post-fertilization) by sampling from the buckets at regular intervals (generally every 10–15 min). Each cross achieved the 4-cell stage at slightly different times, so we adjusted the time of first sampling, interval between sampling, and total sample duration to match the embryos' developmental progress in each cross.

At each sampling point, ~1000 embryos were concentrated onto a 25 μm filter, then transferred in a small volume of culture water to a 1.5 ml centrifuge tube containing 8% paraformaldehyde (final concentration 2% paraformaldehyde in seawater). The preserved embryos were suspended for analysis; three 100 μl aliquots were transferred to a Sedgewick Rafter cell for scoring to their latest completed developmental stage (8- or 16-cells) under a compound microscope.

A logistic curve was fitted for each cross and pCO₂ treatment using R. This curve can be written:

$$p_t = \frac{1}{1 + \exp(-b(t - t_{50}))},$$

where p_t is the predicted proportion of embryos developed (to the ≥ 8 -cell or ≥ 16 -cell stages) at time t (minutes). Parameter b measures the steepness of the logistic curve, and parameter t_{50} is the time at which 50% of the embryos have developed to that stage (gray line on Fig. 1a). For each individual cross, the curve was fitted to all three aliquots, and a quasi-binomial correction was applied to allow for overdispersion due to between-aliquot variability. A within-study meta-analysis could not be conducted with this data set as t_{50} is a single value for each pCO₂ treatment and cross, without a standard deviation. Instead, likelihood ratio tests (corrected for overdispersion) were performed in R to assess the significance of pCO₂ treatment.

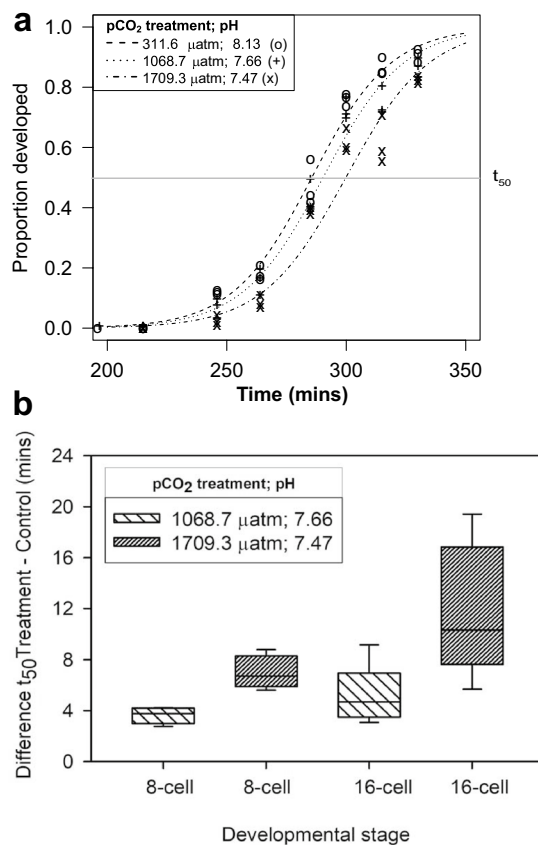


Fig. 1 Early development of *E. chloroticus*. **a** Cross A at the 16-cell stage, as a representative example of the developmental delay in high pCO₂ treatments. Different symbols represent three technical replicates from each culture bucket at each time point; the line shows the logistic curve. t_{50} is the time at which 50% of the embryos have developed to that stage. **b** The difference in t_{50} (minutes) between pCO₂ treatment and Control ($N=5$ crosses) for the 8-cell and 16-cell stages

Part 2: Larval morphometrics (48 h)

Offspring of thirteen single male:female crosses (crosses F to R) were reared in 20 L buckets equilibrated to experimental pCO₂ as described above. Emission scenarios RCP 4.5 and RCP 8.5 at 2100 were used in two separate trials (Part 2a: crosses F, G, H, I; crosses J, K, L, M), and one trial of emission scenarios RCP 8.5 at 2100 and RCP 8.5 at 2150 (Part 2b: Crosses N, O, P, Q, R). Larvae were cultured in OA conditions until 48 h, samples from each bucket were concentrated onto a 25 μ m filter, then transferred to a 1.5 ml centrifuge tube containing 8% paraformaldehyde (final concentration 2% paraformaldehyde in seawater). To measure size, larvae were transferred to a depression slide and photographed using a digital camera (Leica DC500) on a Leica DMRE upright fluorescence microscope. To detect “larval stunting” in OA conditions we measured post-oral arm length on both the left and right side, larval arm width (from the tip of the left post-oral arm to the tip of the right post-oral arm), and body length at midline. We also measured arm angle (angle between skeletal rods of left and right post-oral arms, Fig. 2d) as Chan et al. (2011) showed a change in arm elevation angle in larval *Dendroaster excentricus* under OA conditions. Measurements were made from 10 larvae per treatment from each cross using ImageJ software.

Multivariate analyses of larval morphometrics were conducted for each set of experiments separately (Part 2a: crosses F, G, H, I; crosses J, K, L, M; Part 2b: Crosses N, O, P, Q, R) using PRIMER 7.0.13 (Primer-e, Quest Research Ltd) on an untransformed Euclidean distance matrix. An initial two-way analysis of variance was performed using the PERMANOVA add-in with both pCO₂ treatment and cross as fixed factors (permutation of residuals under a reduced model, 9999 permutations). When required, a one-way analysis of variance for each cross separately (factor pCO₂ fixed) was performed on the same Euclidean distance matrix with the same PERMANOVA settings. The PERMANOVA subroutine “Distance among centroids” and a metric MDS (mMDS) was used to determine the position of the centroid for the 10 measured larvae in multivariate space for each pCO₂ treatment and cross combination using the same Euclidean distance matrix.

Part 3: Metabolic rate

Metabolic rate (pmol O₂ individual⁻¹ h⁻¹) was measured in seven single male:female crosses (Crosses S to Y) with siblings raised in Control or emission scenarios RCP 8.5 at 2100 and RCP 8.5 at 2150. Rates of oxygen consumption of 72 h or 96 h *Evechinus* larvae were measured using a modified flow-through cell (FTC) with an integrated fluorescent oxygen sensor (FTC-PSt3, PreSensTM, Regensburg, Germany) and an Oxy-4 Mini (Presens, Regensburg, Germany),

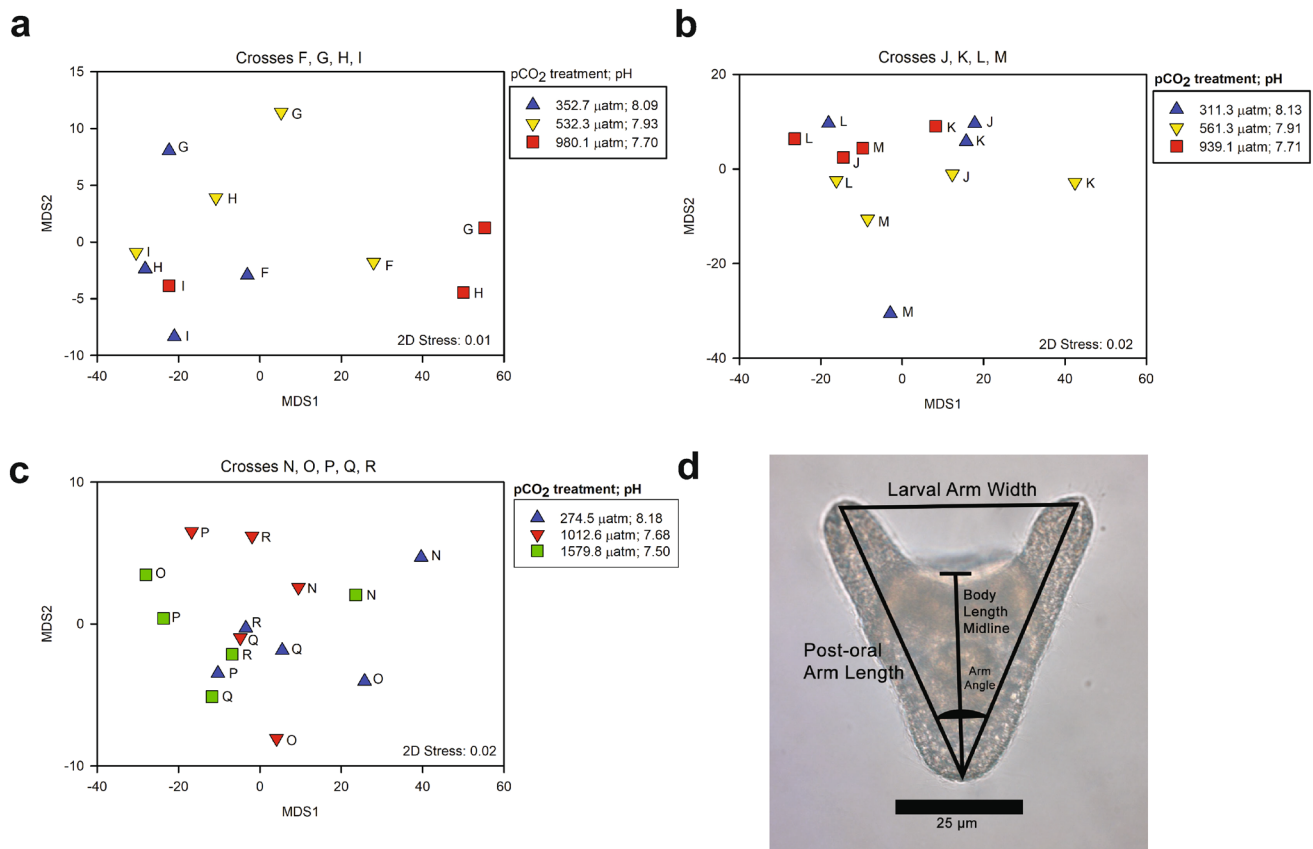


Fig. 2 Effect of culture in elevated CO_2 on 48 h *E. chloroticus* larval morphometrics. Metric MDS (mMDS) plots of centroids **a** Crosses F, G, H, I, at Control, RCP 4.5 and RCP 8.5 at 2100 **b** J, K, L, M, at Control, RCP 4.5 and RCP 8.5 at 2100 and **c** Crosses N, O, P, Q, R at

Control, RCP 8.5 at 2100 and RCP 8.5 at 2150. Exact OA conditions for **a** to **c** (μatm pCO_2 ; pH) are given in legends. **d** 48 h *E. chloroticus* larva showing morphometric measures. Larva from Cross Q, Emission scenario RCP 4.5 2100. Scale bar = 25 μm

as described by Baker et al. (2018). Briefly, small numbers of larvae (20–40) are contained within a static FTC (volume ~ 100 μl) with a 40 μm Nitex filter at either end containing Control or CO_2 -treated seawater. Each respiration run (4 FTCs; 20–30 min) was sibling-matched, with two replicate FTC of the Control and two replicate FTC of one of the treatments (RCP 4.5, RCP 8.5) run for a single cross.

Rates of O_2 consumption were calculated from slopes of the best fit linear regression between % O_2 saturation and time after a 5 min equilibration period. The measurement period was dependent on the rate of O_2 depletion, but O_2 levels were never allowed to drop below 80% saturation. Experimental runs to compare treatments were retained if r^2 exceeded a threshold of $> 90\%$. FTC were calibrated before each use with 100% air saturated filtered seawater and sodium sulphite (20 g/L in deionized water) as a 0% dissolved oxygen standard. Further details on loading and cleaning the FTC between trials are given in Baker et al. (2018).

In a preliminary study, we explored how OA might be affecting mitochondrial function in whole *E. chloroticus*

larvae using two Oroboros O2k Oxygraphs (Oroboros Instruments, Innsbruck, Austria). Each Oxygraph has paired 2 ml respiratory chambers that can simultaneously measure sibling-matched larvae from two different OA treatments. We used two male:female crosses at 72 h post-fertilization: Cross Q at Control and emission scenario RCP 8.5 at 2100, and Cross R at Control and RCP 8.5 at 2150 as in the metabolic experiments described above.

The oxygen electrodes within the Oxygraphs were first calibrated with 100% saturated FSW. Then ca 2000 *E. chloroticus* larvae were transferred by 5 ml pipette into each of the two respirometry chambers (control or pCO_2 -treated waters), and the chamber was allowed to stabilize. To keep the solution homogeneous and prevent damage to the larvae, we used a triangular mixing bar instead of the standard cylindrical mixing bar. Although larval density was high in the Oxygraph chamber during these short-term experiments, respiration rates for *E. chloroticus* larvae were similar when measured in the Oxygraph or in our lower-density FTC described above (see later Results).

Four (4) consecutive measurements were then made on each group of larvae: (1) Routine metabolic rate (RMR) measured as oxygen flux per individual ($\text{pmol O}_2 \text{ individual}^{-1} \text{ h}^{-1}$). (2) Following this, an aliquot of oligomycin ($2.5 \mu\text{g/ml}$) was added to inhibit ATP-synthase inducing State IV⁺, and the oxygen flux was measured when stabilized. This state provides an estimate of proton leak rate, i.e., a measure of oxygen consumption that may not contribute to ATP synthesis, and can present insights to mitochondrial efficiency (LEAK). (3) To measure the electron transport system's maximum oxygen flux capacity, mitochondrial respiration was uncoupled by carbonylcyanide-*p*-(trifluoromethyl) phenylhydrazone (FCCP, $0.25 \mu\text{M}$) by titrating in $0.1 \mu\text{l}$ aliquots until maximal oxygen flux was maintained, without obvious inhibition. This measurement estimates maximal metabolic rate (MMR). (4) To determine background (non-mitochondrial) oxygen flux, we added a final aliquot of rotenone and antimycin A ($1 \mu\text{M}$) to inhibit complexes I and III, respectively. The residual oxygen flux was subtracted from all flux values. Concentrations of the inhibitors (oligomycin, rotenone, antimycin A) followed standard protocols for the instrument (Gnaiger 2020). To avoid hypoxia-induced responses, all experiments were completed with oxygen saturation above 80%. Respiration rates were measured as oxygen flux per individual ($\text{pmol O}_2 \text{ individual}^{-1} \text{ h}^{-1}$), and respiratory flux (JO_2) was calculated in real time as the negative time derivative of the oxygen concentration using Oroboros DatLab Software V 4.1.1.84 (Oroboros Instruments, Innsbruck, Austria).

Statistical approach

Within-study meta-analyses (Garamszegi 2006; Nakagawa and Hauber 2010; Nakagawa and Santos 2012) were used to compare effect sizes between single male:female crosses in the OA experiments in Parts 2 and 3. This approach could not be used in the developmental delay experiments (Part 1) as t_{50} is a single measure without a standard deviation, so a 95% confidence interval could not be calculated.

Effect size was based on Log response ratios (LnRR; Hedges et al. 1999): the ratio of the mean of each treatment (e.g. RCP 4.5 at 2100) was compared to the control for each morphological measure or metabolic rate. Mean effect sizes and 95% confidence intervals were calculated for each cross using OpenMEE (Wallace et al. 2017), which is a GUI for the stable and well-documented meta-analysis R package metafor (Viechtbauer 2010). Meta-analyses were conducted using fixed-effects modeling (simple inverse variance weighting) which provides a conditional inference about the set of k studies included in the meta-analysis (Viechtbauer 2010). Symbol size in the resulting Forest Plot indicates the inverse of the within-study sampling variance, and thus the study weight (smaller symbols indicate higher variances and

smaller weights); the grand mean from the crosses (overall) is indicated by the blue diamond at the bottom of the plot (Wallace et al. 2017). The zero line indicates no effect; negative and positive values for the mean LnRR represent a negative or positive effect of OA, respectively. Individual crosses show significant mean effects when the $\pm 95\%$ confidence interval does not overlap zero. The significance of the overall effect is shown by the p -value adjacent to X-axis.

Results

Seawater chemistry

The carbonate parameters of the seawater used in Parts 1–3 are shown in Table 1. Although we did not achieve the desired $380 \mu\text{atm pCO}_2$ for our control conditions, we were able to attain pCO_2 levels within the predicted range for reduced RCP 4.5 at 2100 (range $500\text{--}650 \mu\text{atm}$), and business as usual emission scenarios of RCP 8.5 at 2100 (range $851\text{--}1370 \mu\text{atm}$), and RCP 8.5 at 2150 (range $1371\text{--}2900 \mu\text{atm}$; IPCC 2014). Across experimental Parts 1–3 there was a consistent reduction in seawater pH, with a decrease of pH of grand mean -0.19 in RCP 4.5 at 2100, -0.47 units in RCP 8.5 at 2100, and grand mean -0.67 units in RCP 8.5 at 2150 (Table 1). Seawaters were not undersaturated for aragonite ($\Omega_a > 1$), except for waters at RCP 8.5 at 2150 with measured $\text{pCO}_2 > 1500 \mu\text{atm}$ (Table 1).

Part 1. Development time to 16-cell stage

E. chloroticus embryos began reaching the 8-cell stage approximately 3 h after fertilization, and the first 16-cell stages were seen after ~ 3.75 h. There were slight differences in timing between the five crosses in control conditions (grand mean Crosses A, B, C, D, E: ≥ 8 -cells $t_{50} = 230.10$ min, range $199\text{--}251$ min, Table 2, Online Resource 1), but the cleavage time (difference between t_{50} at 8-cells and 16-cells) was similar in all crosses (mean $\pm$ SE 55.24 ± 1.71 min). Sibling embryos raised in 1068.9 and $1709.3 \mu\text{atm pCO}_2$ (pH 7.7 and 7.5, respectively) showed a slight developmental delay in all crosses (Fig. 1a), with a mean developmental delay at the 8-cell stage of 3.63 and 7.02 min, respectively (Fig. 1b, Online Resource 1). Based on a 55.24 min cleavage time, this delay corresponds to a treatment:control ratio of 1.066 and 1.127 for 1068.9 and $1709.3 \mu\text{atm pCO}_2$. A greater difference in t_{50} was seen between pCO_2 treatments at the 16-cell stage: at pH 7.7, the mean difference in t_{50} from Control was 5.11 min, while at pH 7.5, the mean difference was 11.85 min (Fig. 1b; Table 2, Online Resource 1). Cross D, in particular, showed the largest change in t_{50} with pCO_2 treatment, having the highest delay at the 8-cell stage (4.22 at pH 7.7 and 8.80 min at

Table 1 Grand means (mean $\pm$ SE) of carbonate chemistry for control and ocean acidification treatments for experimental Parts 1–3

Crosses	Treatment	Measured			Calculated			
		Temp (°C)	Salinity	pH _{TOTAL}	pCO ₂ (μatm)	Ω _c	Ω _a	ΔpH
Part 1: 8 to 16-cell								
A, B, C, D, E	Control	17.8±0.12	33.8±0.32	8.13±0.010	311.6±5.5	4.64±0.07	2.99±0.05	
(N=5)	RCP 8.5 2100	17.8±0.12	33.9±0.32	7.66±0.014	1068.7±38.7	1.84±0.06	1.19±0.04	− 0.47
	RCP 8.5 2150	17.8±0.12	33.6±0.65	7.47±0.015	1709.3±59.8	1.22±0.05	0.79±0.03	− 0.66
Part 2: Larval morphology								
Part 2a								
F, G, H, I	Control	17.8±0.03	34.3±0.04	8.09±0.015	352.7±15.4	4.32±0.12	2.79±0.08	
(N=8)	RCP 4.5 2100	17.9±0.05	34.4±0.06	7.93±0.016	532.3±23.1	3.24±0.41	2.09±0.06	− 0.15
	RCP 8.5 2100	17.8±0.05	34.3±0.05	7.70±0.012	980.1±28.7	1.99±0.05	1.29±0.03	− 0.39
J, K, L, M	Control	18.0±0.08	34.6±0.03	8.13±0.010	311.3±9.1	4.71±0.08	3.04±0.05	
(N=8)	RCP 4.5 2100	18.0±0.06	34.6±0.03	7.91±0.012	561.3±17.4	3.12±0.07	2.02±0.05	− 0.22
	RCP 8.5 2100	17.9±0.09	34.7±0.06	7.71±0.008	939.1±17.8	2.07±0.03	1.33±0.02	− 0.42
Part 2b								
N, O, P, Q, R,	Control	17.6±0.04	34.6±0.06	8.18±0.019	274.5±15.4	5.07±0.17	3.27±0.11	
(N=8)	RCP 8.5 2100	17.7±0.05	34.7±0.05	7.68±0.011	1012.6±27.3	1.93±0.04	1.25±0.03	− 0.48
	RCP 8.5 2150	17.7±0.06	34.6±0.07	7.50±0.007	1579.8±27.1	1.31±0.02	0.85±0.01	− 0.64
Part 3: Metabolic rate								
Part 3a								
S, T, U, V, W, X, Y	Control	17.9±0.05	34.6±0.06	8.14±0.013	308.7±10.5	4.99±0.11	3.23±0.07	
(N=7 for Control, N=6 for	RCP 8.5 2100	17.9±0.07	34.7±0.06	7.65±0.007	1111.1±20.1	1.91±0.03	1.24±0.02	− 0.49
RCP 8.5 2100, RCP 8.5								
2150)	RCP 8.5 2150	17.8±0.04	34.7±0.08	7.46±0.013	1765.5±55.0	1.28±0.03	0.83±0.02	− 0.67
Part 3b								
Q	Control	17.5±0.00	34.6±0.20	8.21±0.016	246.7±11.2	5.36±0.14	3.46±0.09	
(N=2)	RCP 8.5 2100	17.5±0.00	34.8±0.05	7.66±0.030	1076.5±78.2	1.83±0.11	1.18±0.07	− 0.54
R	Control	17.5±0.00	34.7±0.10	8.21±0.047	251.9±34.0	5.32±0.43	3.43±0.28	
(N=2)	RCP 8.5 2150	17.7±0.00	34.8±0.00	7.50±0.009	1572.7±32.9	1.32±0.02	0.85±0.02	− 0.70

Ocean acidification treatments included RCP4.5 at 2100 (500–650 μ atm pCO₂), RCP8.5 at 2100 (851 to 1370 μ atm pCO₂), and RCP8.5 at 2150 (1371 to 2900 μ atm pCO₂; values from IPCC 2014). Full information for each separate Cross is provided in accompanying data file. Calculated parameters (CO2SYS v2.1) used a measured AT value of 2243.0 $\pm$ 3.0 μ mol kg^{−1} (SE, N=55) for Part 1 (Development) and Part 2 (Larval Morphology) experiments in 2012 and 2319.6 $\pm$ 9.3 μ mol kg^{−1} (SE, N=12) for the Part 3 (Metabolic Experiments) in 2013

pH 7.5), with this difference increasing at the 16-cell stage (9.18 at pH 7.7 and 19.43 min at pH 7.5; Table 2, Online Resource 1).

At the 8-cell stage, there was a statistically significant ($p < 0.01$) increase in t_{50} with pCO₂ treatment for all but Cross C (Table 2). At the 16-cell stage, significantly longer t_{50} values ($p < 0.000001$, Table 2) were seen with pCO₂ treatment in all crosses.

Part 2. Larval morphometrics

The impact of OA on larval morphology in *E. chloroticus* was assessed in two sets of experiments: Part 2a which tested emission scenarios RCP 4.5 and RCP 8.5 at 2100 (pH treatments 7.9 and 7.7, Table 1), and Part 2b which tested

emission scenarios RCP 8.5 at 2100 and 2150 (pH treatments 7.7 and 7.5, Table 1).

Three larval measurements were used in multivariate analyses, average arm length, body length, and arm angle (Fig. 2D). Larval arm width was removed as a variable due to high positive correlations with both body length ($r = 0.664$) and average arm length ($r = 0.682$; based on a combined dataset from crosses F, G, H, I, and J, K, L, M). Two-way PERMANOVA on these three larval measurements (arm length, body length, angle) showed a significant effect of pCO₂, cross, and the pCO₂*cross interaction term in all trials (Table 3A). The significant interaction term is also evident in metric MDS plots of the centroid of each pCO₂/cross combination: for example, Cross I at 980.10 μ atm pCO₂ (pH 7.7) shows a position at the opposite end of

Table 2 Time at which 50% of the embryos have developed to the 8-cell or 16-cell stage of development (t_{50}) based on a logistic curve calculated from $N=3$ technical replicates from each culture cross (A, B, C, D, E) at Control, and 1068.7 μatm pCO_2 (pH 7.7) and 1709.3 μatm pCO_2 (pH 7.5)

Cross	Treatment			p value
	Control	1068.7 μatm :	1709.3 μatm :	
	t_{50} (mins)	t_{50} (mins)	t_{50} (mins)	
8-cell stage				
A	229.27	232.03	237.06	0.0085
B	245.04	248.80	250.66	0.0065
C	198.99	202.22	205.15	0.5486
D	249.32	253.54	258.12	7.68E-11
E	227.89	232.07	234.62	0.0017
Mean $\pm$ SE	230.10 $\pm$ 8.85	233.73 $\pm$ 9.00	237.12 $\pm$ 9.09	
16-cell stage				
A	285.58	290.29	299.83	4.80E-14
B	299.66	304.33	309.97	3.24E-19
C	249.05	252.95	258.62	3.99E-07
D	309.99	319.17	329.42	1.72E-36
E	282.44	285.52	288.12	2.28E-10
Mean $\pm$ SE	285.34 $\pm$ 10.34	290.45 $\pm$ 11.06	297.19 $\pm$ 11.79	

The p value is from a likelihood ratio test of the null hypothesis of no pCO_2 effect. Significant p -values shown in bold

MDS Axis 1 than Crosses G and H (Fig. 2a). Because of the significant interaction term we examined differences in the pCO_2 treatment for crosses separately (Table 3B). Significant effects of pCO_2 on multivariate larval morphometrics were seen in 6/8 crosses in Part 2a (crosses F to M) and only 2/5 crosses in Part 2b (crosses N to R) with the high pCO_2 treatment at 1579.8 μatm pCO_2 , pH 7.5 (Table 3B).

Examination of average arm length, body length, and arm angle as univariate measures showed high variability in response to OA between crosses (Fig. 3). Within-study meta-analyses using the crosses as biological replicates shows significant stunting of larval arm and body length at RCP 8.5 at 2100 (pH 7.7) but not at RCP 4.5 at 2100 (pH 7.9). Part 2a: overall effect size includes zero, $p=0.097$, Fig. 4a, b). Larvae cultured at pH 7.9 did, however, show a significant positive effect of arm angle (Fig. 4c). Larvae in Part 2b cultured in seawater at RCP 8.5 at 2100 (pH 7.7) and 2150 (pH 7.5), showed a significant overall negative effect size for arm length, body length and angle at 1579.8 μatm (pH 7.5). Only arm length was significant at 1012 μatm pCO_2 (pH 7.7, Fig. 5), as seen in Part 2a at the same pH (Fig. 4).

To compare the degree of larval stunting in *E. chloroticus* larvae we back-transformed the overall effect size when there was a significant overall effect in the meta-analyses for arm length. Part 2a (crosses G to M) at pH 7.7 had an overall effect size of -0.132 (Fig. 4a), thus the ratio of treatment:control larvae was 0.876; expressed as a percentage, these larvae are 12.4% smaller in arm length. Part 2b (crosses N to R) had an overall effect size of -0.081 at both pH 7.7 and pH 7.5 (Fig. 5a), corresponding to a treatment:control ratio of 0.922 in both OA treatments

(larvae are 7.8% smaller). OA experiments at pH 7.7 in Parts 2a and 2b, thus show a mean treatment:control ratio of 0.899 (based on 0.876, 0.922), with larvae grown in these conditions having 10.1% smaller arm lengths than controls.

Significant differences in body length were seen in meta-analyses for Part 2a (crosses G to M) at pH 7.7 (overall effect size -0.111 ; Fig. 4b) and Part 2b (crosses N to R) at pH 7.5 (overall effect size -0.051 ; Fig. 5b). Back transformation results in a treatment:control ratio of 0.895 at pH 7.7 and 0.950 at pH 7.5, with larvae being 10.5% and 5% smaller in body length, respectively. Inconsistent patterns were seen with body angle, which was significant in Part 2a (crosses G to M) at pH 7.9 (overall effect size 0.055; Fig. 4c) and Part 2b (crosses N to R) at pH 7.5 (overall effect size 0.061; Fig. 5c), resulting in a larger angle in some high CO_2 cultures (treatment:control ratio of 1.057 and 1.063, respectively).

Overall, even though there was high variability in larval measures both within and between crosses (Figs. 2, 3, 4, 5), *E. chloroticus* larvae tend to be ~ 5 –10% smaller in body length and 8–12% smaller in arm length when cultured in OA conditions.

Part 3: Metabolic rate

Metabolic rate measurements in *E. chloroticus* larvae at 72 and 96 h were highly variable across all crosses and pCO_2 treatments (Part 3a), evident in the size of the bars in the box plots (Fig. 6a, b) and in values of the coefficient of variation (SD/mean) which averaged 29% across all cross/ pCO_2 treatment combinations (range 11–51%). Sibling larvae cultured

Table 3 Results of PERMANOVA for Parts 2a (Control, RCP 4.5 at 2100, RCP 8.5 2100) and 2b (Control, RCP 8.5 at 2100, RCP 8.5 2150) to examine the effect of pCO₂ on larval morphometrics

A. Results of two-way PERMANOVA

Cross	pCO ₂		Cross		pCO ₂ *Cross	
	PseudoF	<i>p</i>	PseudoF	<i>p</i>	PseudoF	<i>p</i>
F, G, H, I	PseudoF _{2,99} = 64.472	0.0001	PseudoF _{3,99} = 35.001	0.0001	PseudoF _{5,99} = 12.843	0.0001
J, K, L, M	PseudoF _{2,108} = 8.984	0.0001	PseudoF _{3,108} = 22.304	0.0001	PseudoF _{6,108} = 3.860	0.0001
N, O, P, Q, R	PseudoF _{2,135} = 17.017	0.0001	PseudoF _{4,135} = 21.292	0.0001	PseudoF _{5,99} = 4.589	0.0001

B. Results of one-way PERMANOVA separately for each cross

Cross	Main test		Paired Test					
	Pseudo-F	P(perm)	t	P(perm)	t	P(perm)	t	P(perm)
	F, G, H, I pCO ₂		Control vs RCP4.5		Control vs RCP8.5 at 2100		RCP4.5 vs RCP8.5 at 2100	
F	Pseudo-F _{1,18} = 11.318	0.0008	3.364	0.0017	*	*	*	*
G	Pseudo-F _{2,27} = 38.618	0.0001	3.007	0.0031	9.562	0.0001	5.303	0.0001
H	Pseudo-F _{2,27} = 44.637	0.0001	2.011	0.0199	10.614	0.0001	6.475	0.0001
I	Pseudo-F _{2,27} = 1.422	0.239						
	J, K, L, M pCO ₂		Control vs RCP4.5		Control vs RCP8.5 at 2100		RCP4.5 vs RCP8.5 at 2100	
J	Pseudo-F _{2,27} = 5.195	0.0003	1.159	0.2791	2.781	0.0001	2.324	0.0026
K	Pseudo-F _{2,27} = 9.345	0.0001	3.552	0.0002	0.923	0.4419	3.828	0.0001
L	Pseudo-F _{2,27} = 2.309	0.0644						
M	Pseudo-F _{2,27} = 4.154	0.0083	1.530	0.1072	2.735	0.0039	1.509	0.1079
	N, O, P, Q, R pCO ₂		Control vs RCP8.5 at 2100		Control vs RCP8.5 at 2150		RCP8.5 at 2100 vs RCP8.5 at 2150	
N	Pseudo-F _{2,27} = 9.669	0.0001	3.911	0.0007	2.362	0.0121	2.519	0.0059
O	Pseudo-F _{2,27} = 19.666	0.0001	2.689	0.0033	5.857	0.0001	3.840	0.0008
P	Pseudo-F _{2,27} = 1.788	0.1158						
Q	Pseudo-F _{2,27} = 2.063	0.0888						
R	Pseudo-F _{2,27} = 1.137	0.3376						

Significant results are shown in bold

pCO₂ treatments for Part 2a: 532.3 µatm pCO₂ (pH 7.9) and 980.1 µatm pCO₂ (pH 7.7) for crosses F, G, H, I, and 561.3 µatm pCO₂ (pH 7.9) and 939.1 µatm pCO₂ (pH 7.7) for crosses J, K, L, M; Part 2b: 1012.6 µatm pCO₂ (pH 7.7) and 1579.8 µatm pCO₂ (pH 7.7) for crosses N, O, P, Q, R. (A). Results of two-way PERMANOVA, factors pCO₂ treatment and cross. (B). Results of one-way PERMANOVA (factor: pCO₂ treatment) separately for each cross. *Note Cross F has no pCO₂ treatment at RCP 8.5 at 2100 [980.1 µatm pCO₂ (pH 7.7)] due to a culture crash

at 1111.1 µatm pCO₂ (pH 7.7) had a variable response with cross, with an overall effect size of 0.00 (Fig. 6c) and the 95% CI not including zero for only two crosses—one with a decrease in metabolic rate (Cross U), and one with an increase (Cross V). Culturing at 1765.5 µatm pCO₂ (pH 7.5) generally resulted in a decrease in metabolic rate, with the 95% CI not including zero and negative for three crosses (S, V, W) and positive for Cross T (Fig. 6d). Although the overall effect size was − 0.076, this was also non-significantly different from zero (*p* = 0.242).

Loss of mitochondrial function may be contributing to the depressed metabolic rate of *E. chloroticus* at 1572.7 µatm pCO₂ (pH 7.5) as shown in our preliminary study (Part 3b; Fig. 6e, f). In Cross Q, with 1076.5 µatm pCO₂

(pH 7.7) as the treatment, there was no difference in the RMR, the oligomycin-inhibited LEAK, or the uncoupled MMR between control and treatment (RMR: *t* = − 1.162, *p* = 0.272; LEAK: *t* = − 0.005, *p* = 0.996; MMR: *t* = 0.323, 0.754, *df* = 10, Fig. 6e). However, in Cross R with 1572.7 µatm pCO₂ (pH 7.5), there was a significant depression in RMR (*t* = 2.661, *p* = 0.038, *df* = 6) and MMR (*t* = 3.196, *p* = 0.019, *df* = 6), but not LEAK (*t* = 0.261, *p* = 0.803) between control and treatment.

The ratio MMR:RMR provides information on changes in the capacity of the electron transport system in response to experimental treatment. Larvae reared in control conditions showed an increase in the MMR:RMR ratio (mean ± SE: Cross Q = 1.28 ± 0.16, *N* = 6; Cross R = 1.39 ± 0.08, *N* = 4),

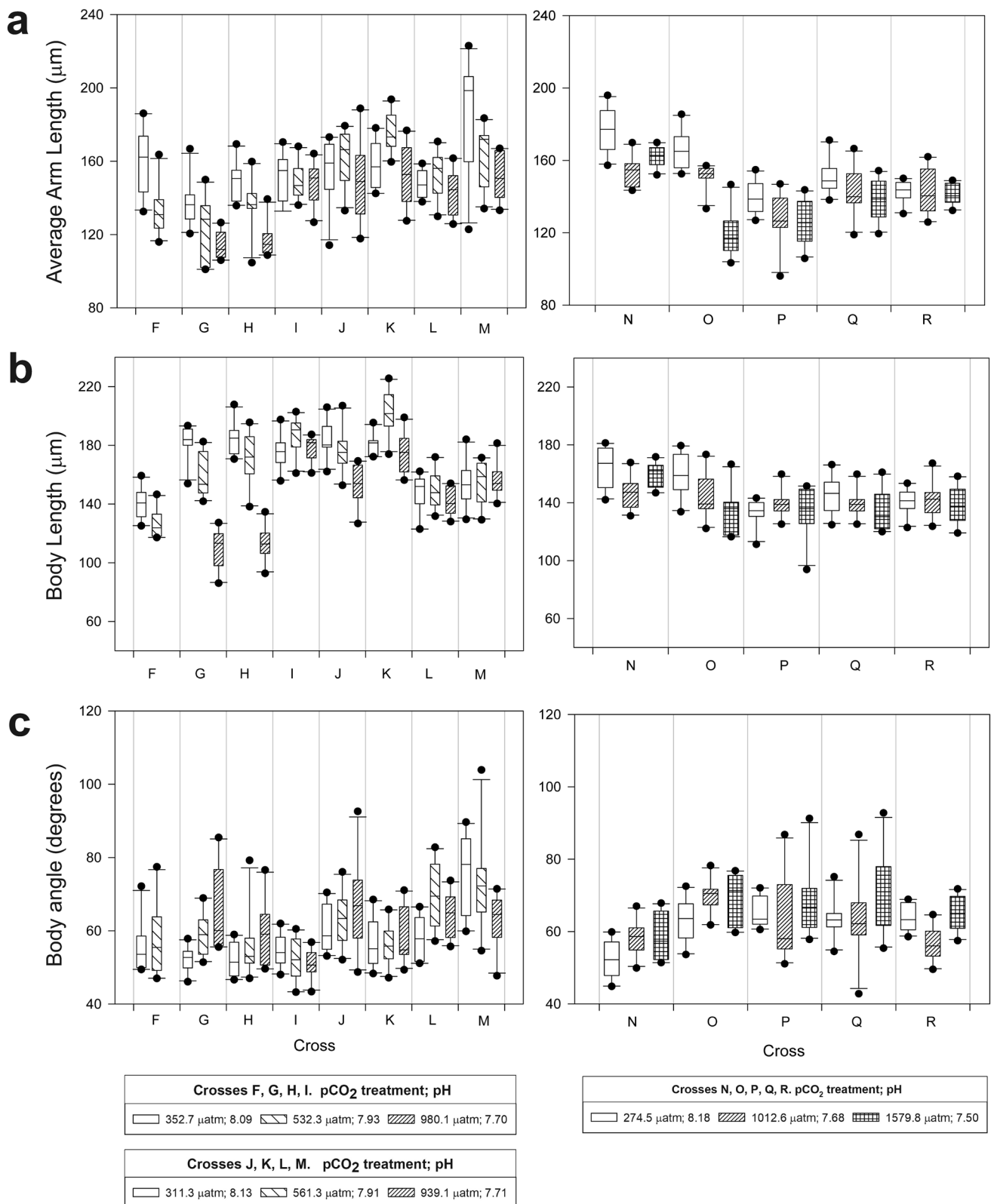


Fig. 3 Boxplots of the larval measures used in the multivariate analyses shown in Fig. 2. **a** Average arm length, **b** Body length, and **c** angle ($N=10$ larvae). The left-hand column includes experiments with RCP 4.5 and RCP 8.5 at 2100 (Crosses F, G, H, I, and J, K, L,

M). The right-hand column includes experiments RCP 8.5 at 2100 and RCP 8.5 at 2150 (Crosses N, O, P, Q, R). Exact OA conditions (μatm pCO_2 ; pH) are given in legends

while those raised in 1076.5 $\mu\text{atm pCO}_2$ had an MMR:RMR ratio close to one (Cross $Q = 1.02 \pm 0.14$, $N = 6$). There was no significant difference in the MMR:RMR ratio between control and 1076.5 $\mu\text{atm pCO}_2$ (pH 7.7) treatment in cross Q ($t = 1.237$, $df = 10$, $p = 0.244$). However, a significant loss of capacity in the electron transport chain was seen in larvae cultured at 1572.7 $\mu\text{atm pCO}_2$ (pH 7.5) with an MMR:RMR ratio of 0.83 ± 0.09 (Cross R, $N = 4$, $t = 3.030$, $df = 6$, $p = 0.0231$).

Discussion

Early-stage *Evechinus chloroticus* were impacted by near-future OA conditions: a developmental delay was seen in the cleavage stages, the larvae had a reduced size at high pCO_2 , and there was a variable response in respiration rate that differed between single male:female crosses. We shall now consider the implications of each Part of this study to the effects of OA on sea urchin larvae.

Part 1. Development to the 8- and 16-cell stage

We used the clear “developmental signposts” (sensu Pörtner et al. 2010) provided by completion of cleavage divisions to show a small, but significant, developmental delay in *E. chloroticus* to the 8- and 16-cell stage at near-future pH (7.7, 7.5); a delay previously shown in sea urchin embryos at more extreme pHs (7.0, Place and Smith, 2012; 6.8, Limatola et al. 2020). Additionally, we found differences in the degree of developmental delay between single male:female crosses, suggesting that the source of the gametes may be important to the magnitude of the delay. While delays of 5–10 min per cleavage may appear minor, there are usually 8–10 cleavages in sea urchins before the blastula stage (Davidson 1989; Bodenstein 2020). A minor delay at each cleavage may have flow-on effects to the total time taken to reach the blastula, and further increase the time to later developmental stages such as gastrula, prism and early echinoplutei. Cumulative small delays may potentially impact larval survival directly, through mortality, or indirectly through longer exposure to predation during the planktonic period (Dorey et al. 2013).

A slowing of development in OA conditions has been previously reported in a range of marine invertebrate species (reviews: Dupont and Thorndyke 2008; Pörtner et al. 2010; Byrne 2012; Przeslawski et al. 2015). The “Guide to best practices for ocean acidification research and data reporting” advocated the use of “developmental signposts” to test for delays in development, noted that quantifying developmental delay was more difficult in species such as echinoderms with semi-continuous development (Pörtner et al. 2010). Studies that have focused on embryo development, where cleavage stages are more discrete,

have been able to quantify delays of ~8 min at first meiotic division in *Strongylocentrotus purpuratus* (pH 7.0, Place and Smith 2012) and ~30 min for cell division in *S. intermedius* (although the % fertilized did not differ at 30 min between Control, pH 7.7, 7.6, and 7.5, Zhan et al. 2016). Delays have also been observed in the time of hatching in blastulae in some studies (pH 7.6, 7.5, Zhan et al. 2016; *Sterechinus neumayeri*, pH 7.9 and 7.8, Yu et al. 2013), but not others (*Echinometra mathaei*, pH 7.9, 7.7, 7.5, Uthicke et al. 2013). However, Yu et al. (2013) suggested that because the embryos inside the fertilization membrane were at a similar developmental stage, there may be a decoupling of the hatching process from other developmental dynamics.

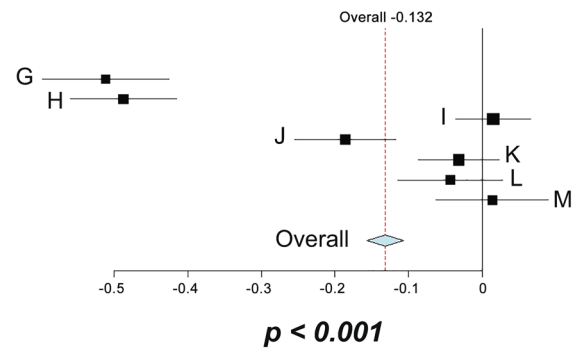
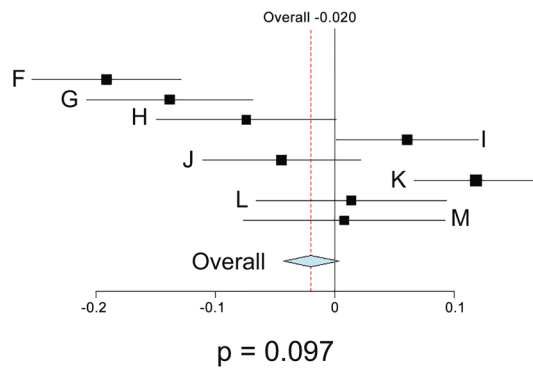
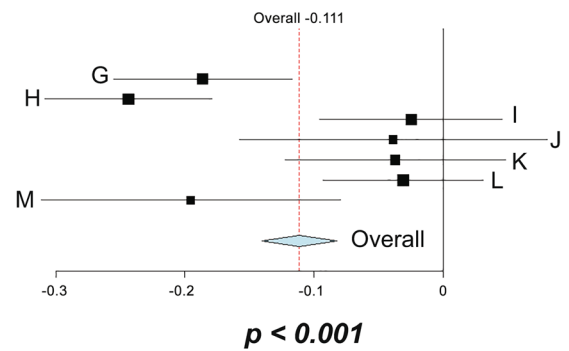
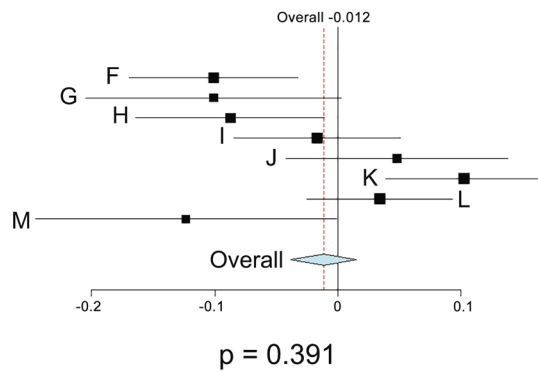
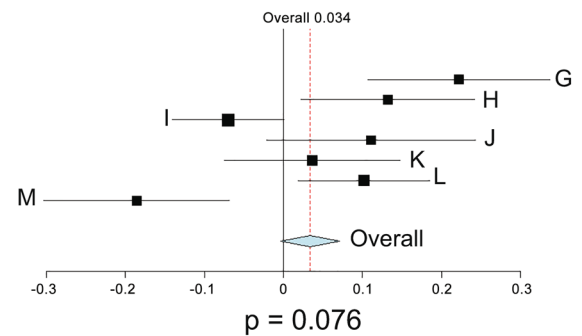
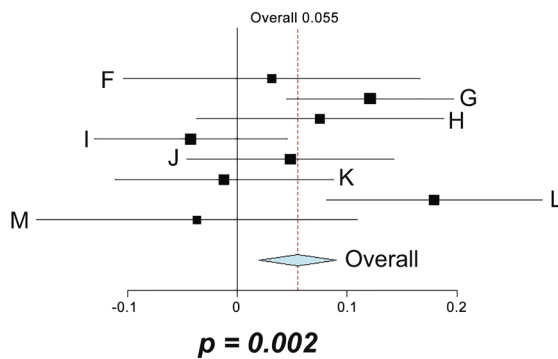
Delay in development to the gastrula and 4-arm echinopluteus has similarly been observed in *Paracentrotus lividus* (pH 7.0, Martin et al. 2011) and in the 4-arm and 6-arm echinoplutei of *S. droebachiensis* over a range of pHs < 7.7 (Dorey et al. 2013). Different results have been reported in *S. purpuratus*: a delay to the 4-arm echinopluteus at pH 7.7 (Stumpp et al. 2011a), but no delay to either the gastrula or 4-arm echinopluteus in a similar study at pH 7.6 by Padilla-Gamiño et al. (2013). However, it is difficult to directly compare studies in *S. purpuratus* as there were also differences in larval feeding (fed: Stumpp et al. 2011a; unfed: Padilla-Gamiño et al. 2013), gamete source (pooled gametes from 3 males/females: Stumpp et al. 2011a; single male:female crosses: Padilla-Gamiño et al. 2013) and, as noted by Padilla-Gamiño et al. (2013), Stumpp et al. (2011a) used morphometric differences in larval skeletons as a measure of developmental delay rather than developmental signposts. Larval size per se does not consider the potential for phenotypic plasticity in larval morphometrics due to the presence of larval food (Strathmann et al. 1992; McAlister and Miner 2018) or the influence of genetic source (Anderson and Hedgecock 2010; Sunday et al. 2011; Kelly et al. 2013).

To better understand the importance of developmental delay in sea urchins we will need to use distinct developmental signposts (e.g. cleavage, and/or distinct embryo stages such as blastula and gastrula), use a sampling interval that can adequately capture what may be only small delays, explore differences in time to developmental signposts between single male:female crosses in both control and OA conditions, and then directly test the impact of embryonic developmental delay by following these crosses through larval development until settlement.

Finally, we need to further explore the mechanism for developmental delay in sea urchins in near-future OA conditions. Place and Smith (2012) tested three cellular biomarkers of cell cycle progression (onset of DNA synthesis, production of the mitotic regulator cyclin B, and formation of the mitotic spindle); none could explain the reduced cleavage rate to the 2-cell stage of *S. purpuratus* at pH 7.0. Using

Control vs RCP 4.5 2100

Control vs RCP 8.5 2100

a Arm Length**b** Body Length**c** Angle

Crosses F, G, H, I. pCO ₂ treatment; pH		
	352.7 μatm; 8.09	 532.3 μatm; 7.93  980.1 μatm; 7.70
Crosses J, K, L, M. pCO ₂ treatment; pH		
	311.3 μatm; 8.13	 561.3 μatm; 7.91  939.1 μatm; 7.71

Fig. 4 Within-study meta-analysis of effect size (LnRR, treatment/control) for 48 h *E. chloroticus* larvae cultured in OA treatments RCP 4.5 at 2100 (left column) and RCP 8.5 at 2100 (right column). Larval morphometrics for **a** average arm length, **b** body length, and **c** angle $N=10$ larvae in Crosses F, G, H, I and J, K, L, M. Forest plot based on the weighted fixed-effect model, with the mean and 95% confidence interval plotted for each cross (cross name shown adjacent to mean $\pm$ 95% CI). The overall mean for all crosses ($N=8$ for RCP 4.5 at 2100; $N=7$ for RCP 8.5 at 2100) is shown by the blue diamond at the bottom of the plot (Overall). The zero line indicates no effect, and a significant difference is seen in crosses where the $\pm$ 95% confidence interval does not include zero. The significance of the overall effect is shown by the p-value adjacent to X axis. The left-hand column compares Control to a treatment of RCP 4.5 at 2100. The right hand column compares Control to a treatment of RCP 8.5 at 2100. Exact OA conditions (μatm pCO₂; pH) are given in figure legend at base. Full OpenMEE output is provided in Online Resource 2

confocal microscopy Limatola et al. (2020) have recently suggested that the delay in embryo development of *Paracentrotus lividus* at pH 6.8 may be a result of changes in the cortical actin cytoskeleton of the egg; although it should be noted that only 16% of the 2 h-old embryos were normal at this pH. Recently, Stumpp et al. (2019) have suggested that developmental delays at low pH result from energy reallocations toward maintenance of cellular pH and biomineralization, leaving less energy available for growth and development (Stumpp et al. 2011a, b; Pan et al. 2015).

Part 2: Larval morphometrics (48 h)

We found significant differences in the body length (5–10% smaller; except in Part 2b: pH 7.7) and arm length (8–12% smaller) of 48 h 4-arm *E. chloroticus* larvae when raised in OA conditions (Part 2a: pH 7.9, 7.7; Part 2b: pH 7.7, 7.5). Clark et al. (2009) conducted OA experiments on *E. chloroticus* at a similar pH (7.7) and found a significant reduction in only one of their morphological measurements (R4, the post-oral skeletal rod), with a treatment:control ratio of 0.783 (extracted from their Fig. 3); so that larvae were 21.7% smaller. Our measure of arm length is equivalent to R1 + R4 of Clark et al. (2009), with a treatment:control ratio of 0.841 at 13 days (cf our arm length ratio of 0.907 at 2 days). However, in their study larvae were raised only from the hatched blastula stage in low pH water, and measurements were made in fed 13-day 4-arm echinoplutei. Consequently, differences in larval age (13 days, 2 days) and feeding (fed, unfed) between the studies make a direct comparison difficult.

During the echinopluteus stage, when growth is primarily through elongation of the larval arms, it is difficult to disentangle the impacts of delay at previous developmental stages, the impacts of lower carbonate saturation on calcification of the skeletal rods, and the importance of larval energetics. We minimized the impact of the latter by sampling non-fed larvae for morphometrics at 48 h, when

maternal lipids were still being used (Sewell 2005); thus, any observed changes in arm length in OA conditions were not confounded by energetic constraints or the presence/absence of food. Nevertheless, we cannot tease apart the important confounding factor of developmental delay from that of larvae raised at high CO₂ being of a smaller size.

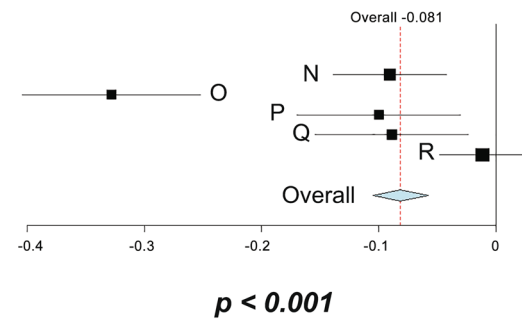
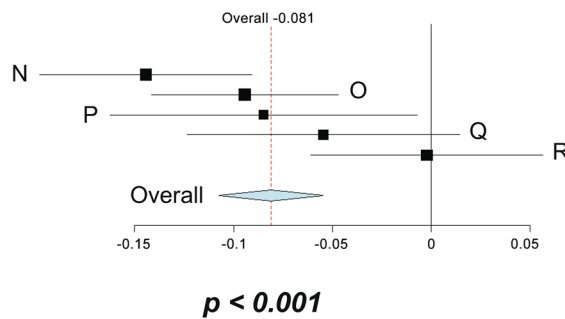
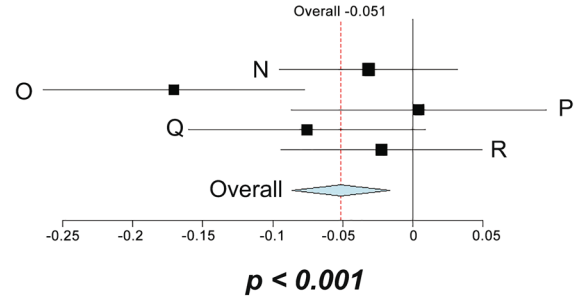
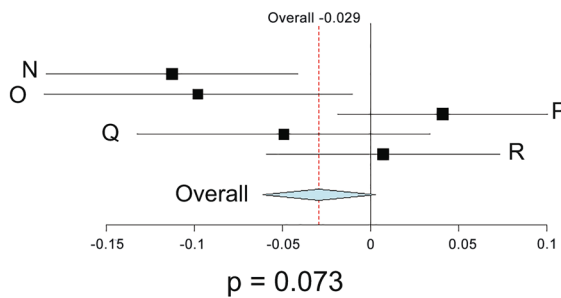
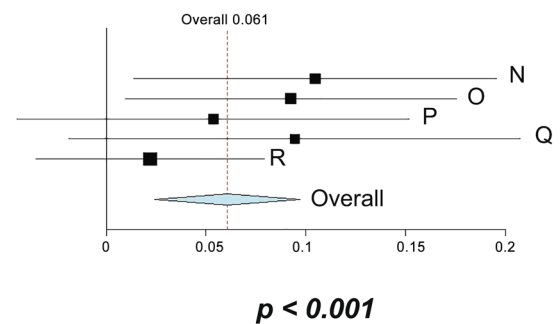
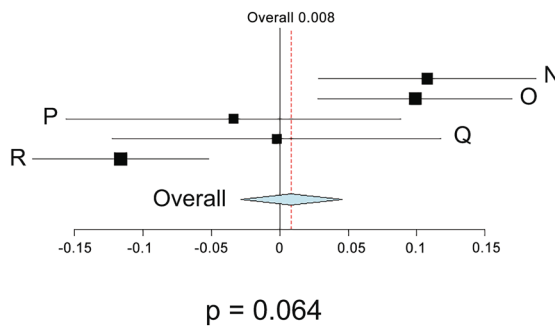
Zhan et al. (2016) is one of the few studies that examines the impact of OA on both embryo development and larval size. They showed a developmental delay at the 16-cell stage in *Strongylocentrotus intermedius*, but because they sampled at specific 30-min time points, the degree of delay is only semi-quantitative. As an index of the delay we calculated at pH. 7.7 that the treatment:control ratio for the percent of fertilized eggs at the 16-cell stage was 0.834 (their Fig. 2F) and 0.861 at the blastula stage (their Fig. 3). In the 4-arm larvae the treatment:control ratio was 0.984 in body rod length and 0.898 in the post-oral rod (their Fig. 5A, C). This suggests that the period between blastula and 4-arm pluteus is a critical period for study to test for the link between embryonic developmental delay and larval size.

Stumpp et al. (2011a, b) reported a developmental delay in *S. purpuratus*, but this estimate was based on the reduced size of the larvae at pH 7.7, rather than by using specific developmental signposts. Using body length as a measure of developmental delay when all larvae were 4-arm echinoplutei (Day 2, Stumpp et al. 2011a, b), the treatment:control ratio has a mean across the three replicate cultures of 0.950, so larvae at pH 7.7 are ~5% smaller (range 2.6–6.8%, calculations from Supplementary Data). We also calculated the sum of Body Rod Length (BRL) and Post-oral arm length (POL) at day 2 to compare to the arm length in *Evechinus* measured here. The treatment: control ratio for arm length (BRL + PO) had a mean of 0.892, although there is considerable variability between replicates (0.782, 0.921, 0.974). Their results suggest that an ~5% developmental delay results in an ~11% decrease in larval size.

Byrne et al. (2013) reviewed the impacts of OA on post-oral arm length from 15 sea urchin species in 22 studies from tropical to polar habitats. In the pH range used in most near-future OA experiments (to ~pH 7.5), the percent decrease in post-oral arm length was generally in the range of 10–20%, though one tropical and polar species showed a percent decrease in the 30–40% band (Byrne et al. 2013). Based on the studies to that time, these authors concluded that this larval stunting was not unequivocally due to a delay in development. The results presented here, and calculated from Zhan et al. (2016) and Stumpp et al. (2011a, b) suggests, however, that simultaneous consideration of both the delay in development and the morphometrics of echinoplutei in multiple single male: female crosses might be a fruitful approach to understanding the mechanisms of reduced larval size in OA conditions.

Control vs RCP 8.5 2100

Control vs RCP 8.5 2150

a Arm Length**b** Body Length**c** Angle

Crosses N, O, P, Q, R. pCO ₂ treatment; pH	
□	274.5 μ atm; 8.18
▨	1012.6 μ atm; 7.68
▩	1579.8 μ atm; 7.50

Part 3: Metabolic rate

Pörtner et al. (2010) suggested that “larval respiration rates represent a comprehensive parameter to characterize

metabolic rates”, and that metabolic measurement might help to unravel the effects of ocean acidification on larvae and larval development (Pörtner et al. (2010, p. 175). Our previous research on *E. chloroticus* showed that the

Fig. 5 Within-study meta-analysis of effect size (LnRR, treatment/control) for 48 h *E. chloroticus* larvae cultured in OA treatments RCP 8.5 at 2100 (left column) and RCP 8.5 at 2150 (right column). Larval morphometrics for **a** average arm length, **b** body length, and **c** angle for $N=10$ larvae in Crosses N, O, P, Q, R. Forest plot based on the weighted fixed-effect model, with the mean and 95% confidence interval plotted for each cross (cross name shown adjacent to mean $\pm$ 95% CI). The overall mean for all crosses ($N=5$ for RCP 8.5 at 2100; $N=5$ for RCP 8.5 at 2150) is shown by the blue diamond at the bottom of the plot (Overall). The zero line indicates no effect, and a significant difference is seen in crosses where the $\pm$ 95% confidence interval does not include zero. The significance of the overall effect is shown by the p-value adjacent to X-axis. The left-hand column compares Control to a treatment of RCP 8.5 at 2100. The right hand column compares Control to a treatment of RCP 8.5 at 2150. Exact OA conditions ($\mu\text{atm pCO}_2$; pH) are given in figure legend at base. Full OpenMEE output is provided in Online Resource 2

respiration rate increased with time post-fertilization, with the rate slowing to an asymptotic level after 48 h (Baker et al. 2018). To reduce the impact of a potential developmental delay under high pCO_2 conditions, we measured respiration rate at 72 or 96 h during the period between 48 and 144 h when all larvae were at the same early 4-arm stage (Sewell et al. 2004). None of the larvae were fed, as metabolic rates of embryos and unfed larvae can be elevated over fed larvae in OA conditions (Pan et al. 2015).

Even with these considerations, intended to reduce the sources of variability, we found evidence for both increases and decreases in metabolic rate under high pCO_2 , depending on the male:female cross (= source of gametes), and with no significant overall effect of OA on routine metabolic rate at either pH 7.7 or pH 7.5. Similar cross differences in larval respiration rate were seen in the New Zealand sea urchins *Centrostephanus rodgersi* and *Fellaster zelandiae* using the same experimental protocols (Baker et al. 2018).

Previous studies on the impacts of OA on respiration rate in sea urchin larvae have shown a variable response, with significant effects at lower pH (Stumpp et al. 2011a, b; Padillo-Gamiño et al. 2013; Gaitán-Espitia et al. 2017; Lee et al. 2019; Karelitz et al. 2019), or no significant effects (Kelly et al. 2013; Pan et al. 2015; Wong and Hofmann 2020, present study). Several factors may be contributing to the different responses to OA, including the sea urchin species, the source populations (Kelly et al. 2013; Gaitán-Espitia et al. 2017), and parental acclimation (Karelitz et al. 2019). Some studies have also noted a higher variability in the respiration rate between replicate cultures at low pH (Stumpp et al. 2011a). In addition, high variability can occur between the replicate sealed vials used to measure respiration, as a measurable signal requires pooling large numbers of larvae (generally > 20), with each larva having an individual response to OA.

Gene expression studies in sea urchins have also provided evidence for changes in metabolic rate in OA conditions (reviews: Evans and Watson-Wynn 2014; Strader

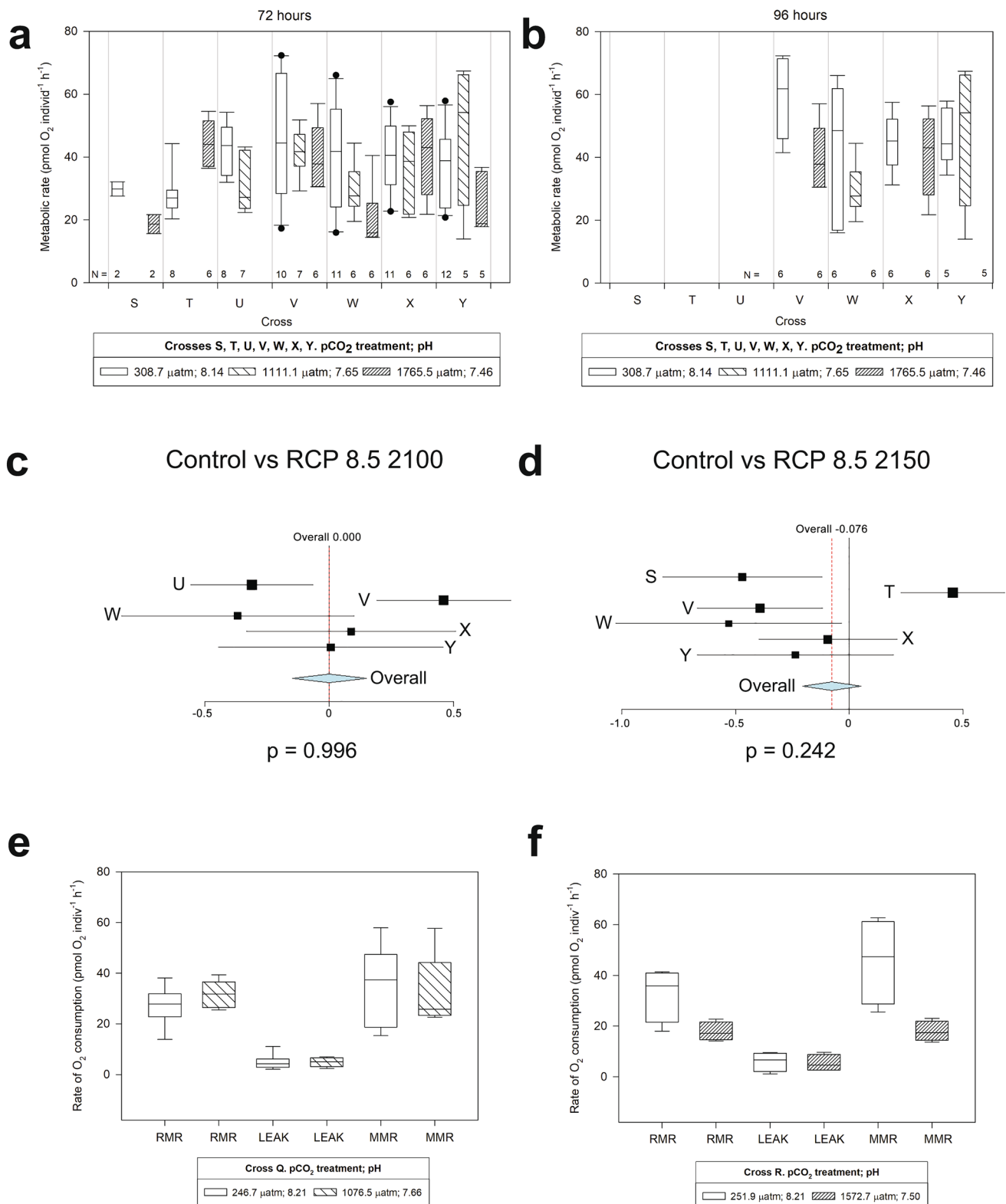
et al. 2020). In particular, high pCO_2 results in a decrease in ATP production by down-regulating the expression of genes involved in the tricarboxylic acid cycle (TCA), electron transport chain, and oxidative phosphorylation that occur within the mitochondria (Evans and Watson-Wynn 2014; Strader et al. 2020). Loss of mitochondrial capacity in *E. chloroticus* sperm (Hudson 2010) and larvae in high pCO_2 (herein) also suggests that some of the changes in metabolic rate under OA conditions may involve impacts of pH on mitochondrial function.

Future directions

Byrne and Hernandez (2020) in a recent review summarized the negative effects of OA on whole sea urchin larvae, including delayed development, altered body morphology, changes to metabolism and reduced scope for growth. Our data on *E. chloroticus* larvae, in general, supports these conclusions—we found negative impacts of OA on early development, with slower developmental times and smaller larvae; although we did not show a significant effect of OA on metabolic rate. Additionally, we have shown considerable variation in all measured physiological responses between different male:female crosses.

Previous research assessing scope for growth in sea urchin larvae suggested that the costs of pH_i regulation in OA conditions could lead to a shift in the energy budget, with less energy available for growth (Stumpp et al. 2011a). This hypothesis was tested by Jager et al. (2016) using Dynamic Energy Budget (DEB) models based on three experimental studies on *Strongylocentrotus droebachiensis* (Dorey et al. 2013; Chan et al. 2015; Stumpp et al. 2013). Interestingly, differences in the DEB parameters in Stumpp et al. (2013), when compared to the other studies (Dorey et al. 2013; Chan et al. 2015), were attributed by Jager et al. (2016) to a smaller egg size, and presumably reduced maternal provisioning, at the start of the experiments.

Two studies, both in *Strongylocentrotus purpuratus*, have directly investigated the importance of egg lipid and protein reserves on the response of embryos and larvae to OA. Matson et al. (2012) found lipid utilization rates and protein content in *S. purpuratus* did not vary with pCO_2 , and Pan et al. (2015) also found OA did not affect whole-body protein content in embryos and larvae (fed or unfed). Matson et al. (2012) suggested that energy utilization might be hard-wired during early development and that this is robust to environmental stress, or that *S. purpuratus* from the collection location (Santa Barbara Channel) with a variable pCO_2 environment had the physiological capacity to resist exposure to moderately high pCO_2 water masses. Future research on biochemical utilization in sea urchin embryos and larvae from a broad range of species and locations under OA conditions is critical.



Teasing apart the impacts of developmental delay, larval energetics, and the “interaction between the stunting impacts of acidification on skeletogenesis and shorter arm lengths in the presence of favorable food conditions is likely to be

complex, but is essential to understand” (Byrne and Hernandez 2020, p. 286). Conducting such OA experiments in sea urchins will require innovative thinking and approaches concerning design and logistics.

Fig. 6 Metabolic rate of *E. chloroticus* larvae. **a** Metabolic rate at 72 h. Sample sizes for box plots are shown above X-axis. **b** Metabolic rate at 96 h. Sample sizes for box plots are shown above X-axis. **c** Within-study meta-analysis of effect size (LnRR, treatment/control) for metabolic rate in 72 and 96 h *E. chloroticus* larvae cultured in elevated CO₂. Forest plot of LnRR comparing Control and RCP 8.5 at 2100 ($N=5$ crosses). **d** Forest plot of LnRR comparing target pCO₂ of Control and RCP 8.5 at 2150 ($N=6$ crosses). For details on Forest plots, see legend for Figs. 4 and 5. **e**. Preliminary experiments on mitochondrial function with Cross Q with Control and 1076.5 μ atm pCO₂ ($N=6$ for all measures). **f** Preliminary experiments on mitochondrial function with Cross R with Control and 1572.7 μ atm pCO₂ ($N=4$ for all measures). RMR=routine metabolic rate [equivalent to metabolic rate measured at 72 h and 96 h as shown in **a** and **b** above]; LEAK=proton leak rate [a measure of oxygen consumption not contributing to ATP synthesis]; MMR=maximum metabolic rate [a measure of the electron transport system's maximum oxygen flux capacity; see Methods for further explanation]

Firstly, we need detailed information on developmental delay using “signposts” through the whole of embryo development until the earliest 4-arm echinoplutei. This will require regular sampling to hit the time of transition to cleavage stages, blastula, gastrula, the timing of formation of the tri-radiate spicule (e.g., Yu et al. 2013), and then through to the prism and 4-arm echinopluteus. Over the first few days of development, this high-intensity sampling will require a dedicated team of researchers, as small changes in timing will be expected between single male:female crosses and due to the OA treatment(s).

Secondly, there needs to be a critical control of the amount of food in the feeding larval stages. Many sea urchins, such as *E. chloroticus* (Sewell et al. 2004), will not develop past the 4-arm stage without algal food. However, as noted by Chan et al. (2018), it is standard in larval experiments to apply the same algal food concentration to both control and treatment conditions, without considering the potential for confounding due to increased mortality in either condition (i.e., the treatment with the highest mortality has a higher algal ration for the survivors). To keep food level constant and avoid phenotypic plasticity in arm length due to variable algal concentrations (Strathmann et al. 1992; McAlister and Miner 2018) would require continuous monitoring of larval density in the culture containers and addition of a consistent number of algal cells/larva at the time of feeding. This is another labor-intensive operation over the 3–4 weeks of larval development, but with the advantage that samples for biochemical analyses to detect changes in energy allocation would then be from larvae receiving the same amount of algal food.

Thirdly, we need a better understanding of whether smaller echinoplutei in OA treatments is due to delayed development (point 1), whether they can “catch up” on the same ration (point 2), or if they are energetically challenged,

impacting the building and maintenance of the calcified skeleton and producing less biomineral mass per stage (Stumpp et al. 2012; Byrne et al. 2013; Byrne and Hernandez 2020). Recent advances in methodologies and techniques to understand pH regulation in calcification of sea urchin embryos and larvae (Stumpp et al. 2012, 2019; Hu et al. 2020) provide the opportunity through training and/or collaborations to develop new approaches to address this.

Finally, the single cross approach used here has emphasized that the offspring of sea urchins collected from the same population can have variable responses to OA. To understand the potential impacts of OA on *Evechinus chloroticus* and other sea urchin species, we need to have a good understanding of the level of variation in physiological responses in both a single population and, then more broadly, throughout the geographic range.

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Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval The authors declare that all applicable national and institutional guidelines for sampling, care and experimental use of organisms were followed in this study.

Consent to participate Not applicable.

Consent for publication Not applicable.

Data availability statement All data generated or analyzed during this study are included in this published article [and its supplementary information files].

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